

3.1 Introduction to determinants

I

Definition

For $n \geq 2$, the determinant of an $n \times n$ matrix $A = [a_{ij}]$ is the sum of n terms of the form $\pm a_{1j}$, with plus and minus signs alternating, where the entries

$a_{11}, a_{12}, \dots, a_{1n}$ are from the first row of A . In symbols;

$$\det A = a_{11} \det A_{11} - a_{12} \det A_{12} + \dots + (-1)^{1+n} a_{1n} \det A_{1n}$$

$$= \sum_{j=1}^n (-1)^{1+j} a_{1j} \det A_{1j}$$

Example 1

Compute the determinant of

$$A = \begin{bmatrix} 1 & 5 & 0 \\ 2 & 4 & -1 \\ 0 & -2 & 0 \end{bmatrix}$$

Solution

$$\det A = a_{11} \det A_{11} - a_{12} \det A_{12} + a_{13} \det A_{13}$$

$$A_{11} = \begin{bmatrix} 4 & -1 \\ -2 & 0 \end{bmatrix}, A_{12} = \begin{bmatrix} 2 & -1 \\ 0 & 0 \end{bmatrix}, A_{13} = \begin{bmatrix} 2 & 4 \\ 0 & -2 \end{bmatrix}$$

$$\det A = 1(0-2) - 5(0-0) + 0(-4-0) = -2$$

Let's choose row 2 instead of 1.

$$\det A = -a_{21} \det A_{21} + a_{22} \det A_{22} - a_{23} \det A_{23}$$

$$A_{21} = \begin{bmatrix} 5 & 0 \\ -2 & 0 \end{bmatrix}, A_{22} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, A_{23} = \begin{bmatrix} 1 & 5 \\ 0 & -2 \end{bmatrix}$$

$$\det A = -2(0-0) + 4(0-0) + 1(-2-0) = -2$$

Be aware of the sign pattern: $\begin{bmatrix} + & - & + \\ - & + & \dots \\ + & - & \dots \end{bmatrix}$

Another common notation is:

$$\det A = 1 \begin{vmatrix} 4 & -1 \\ -2 & 0 \end{vmatrix} - 5 \begin{vmatrix} 2 & -1 \\ 0 & 0 \end{vmatrix} + 0 \begin{vmatrix} 2 & 4 \\ 0 & -2 \end{vmatrix} = -2$$

3.1 Introduction to determinants (cont'd)

II

To state the next theorem, it is convenient to write the definition of $\det A$ in a slightly different form. Given $A = [a_{ij}]$, the (i, j) -cofactor of A is the number C_{ij} given by

$$C_{ij} = (-1)^{i+j} \det A_{ij} \quad (4)$$

Then

$$\det A = a_{11}C_{11} + a_{12}C_{12} + \dots + a_{1n}C_{1n}$$

This is called a cofactor expansion, in this example across the first row. We can choose row at will.

Example 2

Use cofactor expansion across the third row to compute $\det A$, where

$$A = \begin{bmatrix} 1 & 5 & 0 \\ 2 & 4 & -1 \\ 0 & -2 & 0 \end{bmatrix}$$

Theorem 1

The determinant of an $n \times n$ matrix A can be computed by a cofactor expansion across any row or down any column. The expansion across the i th row using the cofactors in (4) is

$$\det A = a_{i1}C_{i1} + a_{i2}C_{i2} + \dots + a_{in}C_{in}$$

The cofactor expansion down the j th column is

$$\det A = a_{1j}C_{1j} + a_{2j}C_{2j} + \dots + a_{nj}C_{nj}$$

Solution

$$\begin{aligned} \det A &= a_{31}C_{31} + a_{32}C_{32} + a_{33}C_{33} \\ &= (-1)^{3+1} a_{31} \det A_{31} + (-1)^{3+2} a_{32} \det A_{32} + (-1)^{3+3} a_{33} \det A_{33} \\ &= 0 \begin{vmatrix} 5 & 0 \\ 4 & -1 \end{vmatrix} - (-2) \begin{vmatrix} 1 & 0 \\ 2 & -1 \end{vmatrix} + 0 \begin{vmatrix} 1 & 5 \\ 2 & 4 \end{vmatrix} \\ &= 0(-2+0) = -2 \end{aligned}$$

Introduction to determinants (cont'd)

III

Example 3

Compute $\det A$ where

$$A = \begin{bmatrix} \textcircled{3} & -7 & 8 & 9 & -6 \\ 0 & 2 & -5 & 7 & 3 \\ 0 & 0 & 1 & 5 & 0 \\ 0 & 0 & 2 & 4 & -1 \\ 0 & 0 & 0 & -2 & 0 \end{bmatrix}$$

Solution

The cofactor expansion down the first column has all but the first term equal to zero.

Thus

$$\det A = 3 \begin{vmatrix} \textcircled{2} & -5 & 7 & 3 \\ 0 & 1 & 5 & 0 \\ 0 & 2 & 4 & -1 \\ 0 & 0 & -2 & 0 \end{vmatrix} + 0C_{21} + 0C_{31} + 0C_{41} + 0C_{51}$$

Next we expand this 4×4 determinant down the first column to take advantage of the zeros there.

$$\det A = 3(2) \begin{vmatrix} 1 & 5 & 0 \\ 2 & 4 & -1 \\ 0 & -2 & 0 \end{vmatrix}$$

Let's now take advantage of the zeros in row three.

$$\det A = 3(2) \left[(0) \begin{vmatrix} 5 & 0 \\ 4 & -1 \end{vmatrix} - (-2) \begin{vmatrix} 1 & 0 \\ 2 & -1 \end{vmatrix} + 0 \begin{vmatrix} 1 & 5 \\ 2 & 4 \end{vmatrix} \right]$$
$$= 6[0 - 2 + 0] = -12$$

3.1. Introduction to determinants (cont'd)

IV

Theorem 2

If A is a triangular matrix, then $\det A$ is the product of the entries on the main diagonal of A .

Practice problem

Compute $\det A$

$$A = \begin{bmatrix} 5 & -7 & 2 & 2 \\ 0 & 3 & 0 & -4 \\ -5 & -8 & 0 & 3 \\ 0 & 5 & 0 & -6 \end{bmatrix}$$

Let's take advantage of all the zeros in column three.

$$\det A = 2 \begin{vmatrix} 0 & 3 & -4 \\ -5 & -8 & 3 \\ 0 & 5 & -6 \end{vmatrix} \rightarrow \text{rest of the terms are zeros}$$

Now let's take advantage of the zeros in the first column. Keep in mind the sign order

$$\begin{vmatrix} + & - & + \\ - & + & - \\ + & - & + \end{vmatrix}$$

We don't write out the zero terms.

$$\begin{aligned} \det A &= 2(-1)(-5) \begin{vmatrix} 3 & -4 \\ 5 & -6 \end{vmatrix} \\ &= 10(3(-6) - 5(-4)) \\ &= 10(-18 + 20) = 20 \end{aligned}$$